American Sign Language Machine Learning based Translation Model

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***Abstract*—Reducing communication barriers between the hearing-impaired community and the broader society remains a critical challenge, as recognizing sign language gestures requires expertise that most people do not possess. This paper addresses this problem by evaluating and designing an ASL recognition system using deep learning approaches. The main objectives are to identify the most accurate models for both static alphabet-level and dynamic word-level gesture recognition, and to incorporate them into a real-time recognition system. For image-based recognition, CNN-based and hybrid models are trained and evaluated on the ASL Alphabet datasets. For video-based recognition, pretrained SlowFast, Slow-only, and ViViT models are evaluated on WLASL100 and WLASL300 subsets, and further tested on a custom-collected set of 300 sign language videos. The main results are as follows: (1) image-based experiments identify ResNet as best-performing model with accuracy >93 % for different configurations ; (2) video-based experiments show ViViT achieving 51.9% and 27.5%, SlowFast achieving 65.38% and 61.38%, and Slow-only achieving 64.34% and 54.24% on WLASL100 and WLASL300 respectively; (3) inference on the custom test set yields the highest accuracy of (61%) with SlowFast Networks. These results confirm that CNN-based approaches remain state-of-the-art for static recognition, while spatiotemporal models are essential for dynamic word-level gesture recognition.**

***Index Terms*—Sign Language Recognition, Convolutional Neural Networks (CNNs), ViViT Model, SlowFast Networks, Slow-only Model, Real-time Recognition**

I. Introduction

Sign language recognition using deep learning techniques is one of the important studies which is directed for improving communication between hearing-impaired groups of people and the rest of society. American Sign Language (ASL) is based on a combination of hand shapes, spatial layout and temporal movements. In this way, automatic recognition of sign language is a complex problem.

With the fast development of deep learning approaches, Convolutional Neural Networks (CNNs) have shown high performance in static gesture recognition. Models like AlexNet and ResNet have been widely used for ASL alphabet classification and reached high accuracy on image datasets [1], [2]. Additionally, Vision Transformers (ViT) and LSTM-based architectures can also be applied on image-based recognition.

However, these models mainly focus on spatial feature extraction and perform well on image-based recognition, as gestures are pertained as independent samples. In real-world cases, sign language communication is based on continuous motion and temporal patterns, which highlight the demand for models that can capture both spatial and temporal details from video inputs.

To address this, latest advancements in deep learning have proposed spatiotemporal models such as SlowFast and Video Vision Transformers (ViViT), which are particularly designed for video recognition tasks. These approaches capture motion patterns and temporal information with high efficiency, which can help to enhance word-level sign recognition processes [3], [4].

Despite the recent progress, a biggest challenge in sign language recognition remains the availability and quality of datasets. Mostly, open-source datasets consist of static images of alphabet gestures. And large-scale video datasets for word-level gestures persist and often contain class imbalance issues. Such limitations become the reason for the gap between efficient models and practical systems capable of recognizing sign language in real-world scenarios.

Taking into consideration these challenges, this project emphasized on a comparative study of deep learning architectures to identify models' accuracies for ASL recognition. And based on this research, identify the best-perfoming models for the real-time recognitions tasks. By handling both static and dynamic ASL recognition, the project is aimed for reducing the gap between simple alphabet datasets and practical communication settings.

The main contributions of this project are as follows:

- Propose two system designs for American Sign Language recognition: image-based (alphabet-level) and video-based (word-level)
- Accomplish a comparative analysis of different deep learning approaches for ASL datasets [5], [6].

- Adapt and utilize pretrained SlowFast and ViViT models on video-based WLASL dataset [7] and evaluate inference on custom test subset.
  Our codes are on GitHub.[1]

The rest of this paper is structured as follows. Some of the constraints which influence our project are discussed in Section II. As for Section III, it represents the background and related work. The project approach which includes solutions and features is described in Section IV. Section V highlights the project execution details including design-making decisions. Section VI contains evaluation of main experimental and practical results. Final Section VII includes possible directions of future work and key findings.

II. Constraints

1) Hardware and Resource constraints:

The major constraint affecting our project execution is hardware availability. Due to lack of access to dedicated research infrastructure, all training and evaluation was done on Kaggle with restricted cloud trial hours and consumer-grade GPUs. This directly limited training duration and batch sizes. For ViViT and SlowFast models design decisions resulted on using Kinetics-400 pre-trained weights, not training from scratch. This limitation defined the scope and boundaries in which evaluation was done.

2) Data and dataset constraints:

Dataset related constraints affect our project the most. Primarily, it is the limited diversity and quality of available datasets. As for image datasets, there are an excessive number of images per class, for faithful model comparison downsampling is conducted. Publicly available video datasets are collected from a limited number of signers, skin tones, lightning and low divergence in signing styles. In this way, our proposed gesture recognition solution can not be considered as generalized.

3) Economic constraints:

This project was conducted without a dedicated budget. Dataset choices and experimental setup relied on open-source and freely available resources. This lead to limitations in assessing sign language datasets with higher quality, duration of cloud GPU training runs, use of dedicated hardware like depth sensors which potentially would enhance gesture capturing. Any future use of this system design for producing deployable system would consider computational and data collection costs.

4) Social and Ethical constraints:

Sign Language Recognition is socially impactful and vital challenge as it aims to reduce communication barriers between different groups. Due to the fact that WLASL dataset was obtained from finite pool of signers, models evaluation can not be counted as generalized across regional ASL variability, signing styles, skin tones and lightning conditions. In this way, considering such constraints at design level, the current performance results and architecture choices should not be defined as general-purpose accessibility utility.

5) Safety and Health constraints:

There are no direct health and safety related limitations, as our evaluation and system design do not interact with users. However, design decisions considered the fact that gestures recognition without sufficient accuracy and high generalization could lead to miscommunication or dangerous consequences in extreme situations. That's why, for system design threshold for accuracy is set and confidence score outputs are shown.

III. Background and related work

A. Background

Sign language recognition is a computer vision problem that is said to interpret visual gestures into a text or semantic form. Sign language video capturing can be classified into two large categories: *image-based (static) and video-based (dynamic) recognition*. In univariate recognition, frames are considered separately and indicate one gesture, e.g. a letter of the alphabet. Conversely, dynamic recognition processes frame sequences, with the meaning being determined by hand shape, but also by time motion, space positioning and gesture-gesture transitions [7].

Sign language recognition entails several processes such as extractions of spatial features, time modeling and recognition. Spatial features are hand shape, orientation, and position which would usually be modeled with Convolutional Neural Networks (CNNs). Temporal characteristics define the movement patterns on a frame to frame basis and are necessary to identify words and phrases. They are trained with sequential or video-based models like Long Short-Term Memory (LSTM) networks or transformer-based models [8].

Formally, given a video sequence $V = \{f_1, f_2, ..., f_T\}$, the recognition process can be described as:

$$y = F(V) \tag{1}$$

where $F(\cdot)$ represents a deep learning model that extracts spatio-temporal features and maps them to a label y, such as a letter or word.

Existing systems typically follow a pipeline consisting of:

- Frame sampling and preprocessing
- Feature extraction (CNN / Transformer)
- Temporal modeling (LSTM / attention / video CNN)

[1] https://github.com/AbdollaZ/seniorproject_signlanguage

- Classification layer

Early models were tailored to solve a static classification task, where CNN-based networks like AlexNet and ResNet were very accurate at recognizing ASL alphabets [9]. The models, however, cannot capture motion information and thus they are restricted to simple tasks. More sophisticated systems make use of the temporal modeling that is required in the recognition of complete words and phrases.

B. Motivation

Although deep learning has made great strides, a single system is yet to be developed that can identify sign language effectively and efficiently in real-life scenarios. Current methods are a trade-off among accuracy, complexity and practicality.

CNN-based models are very effective in recognizing letters at the alphabet level, and the accuracy of these models is about more than 95percent [9]. Nonetheless, these models have users typing word by word, which is slow and not natural in actual communication. Conversely, word-level recognition systems, including those built on the WLASL dataset are trying to capture entire gestures but are less accurate because of motion variability, intersigner variability, and complexity of datasets [7].

The hybrid methods like CNN + LSTM are better at temporal models but still have difficulties with long-range relationships and intricate patterns of movement [8]. In more recent architectures, such as SlowFast networks and Vision Transformers (ViViT), these limitations are overcome by more effectively modeling both spatial and temporal information in a better way. These models are increased in cost of computation and more difficult to implement in real time systems.

Motivating example: In our project, we find out that an ASL letter recognition CNN-based model (AlexNet/ResNet) is very effective and attains a high accuracy because the task is very simple. But when it is used on word-level recognition (WLASL) the same method does not match motion resulting in poorer performance. Video-based models, in contrast, like SlowFast and ViViT, enhance recognition by using temporal information, at the cost of greater computational complexity.

This motivates the need to explore two complementary approaches:

Efficient CNN-based models for static gestures Advanced video models for dynamic word recognition

C. Related Work

Sign language recognition has been applied to numerous deep learning models in the past years. These methods can be divided into three: models of the approach of static CNN: based models, models of hybrid spatio-temporal models, and models of video-based architecture.

1) CNN-based models (static recognition): CNN-based methods have extensively been applied in recognizing ASL alphabets because of their capability to extract spatial features. Gangal et al. showed that CNN models may reach up to 96percent of accuracy on fixed ASL sets of data [9]. Equally, ResNet-based architectures have also been very effective in performance because of their deep feature extraction ability.

These models however, assume that frames are independent and thus they are not able to model motion. Consequently, they can only do static gestures and cannot be used to identify continuous or word-level sign language.

2) Hybrid models (CNN + LSTM): In order to solve the issue of temporal dependencies, various works use CNNs with sequential models. The CNN-LSTM architecture suggested by Kumari et al. involves CNN that draws spatial feature and LSTM that learns temporal relationships [8]. This gives better results on dynamic gestures than pure CNN models.

Similar hybrid CNN–LSTM approaches have been explored in recent studies [8]. But LSTM-based solutions lack long-range dependency modeling and tend to be weak in modeling complex motion patterns.

3) Word-level recognition and datasets: One of the important contributions to the field was the introduction of the WLASL dataset by Li et al. [7]. This data includes more than 2,000 words and shows the difficulties of large-scale sign language recognition, such as high inter-class variability, variability among signers, and motion variations.

The paper revealed that the state-of-the-art CNN models are not effective on large vocabulary tasks, highlighting the importance of improved temporal models.

4) Video-based CNN architectures: Feichtenhofer et al. proposed the SlowFast architecture, which is based on two paths: a slow pathway computing spatial semantics, and a fast pathway computing motion dynamics [4]. This design enables the model to record both the appearance and motion at the same time.

In sign language recognition, Radhakrishnan et al. showed that SlowFast-based models are more effective than conventional CNN approaches in recognizing sign language in general [10].

5) Transformer-based models: Arnab et al. suggested ViViT, which is a transformer architecture that encodes video data with self-attention in both spatial and temporal dimensions ViViT. This enables the model to model long-range dependencies better than CNN-based approaches [3].

Recent studies by Brettmann et al. demonstrated that transformer-based models are superior to CNN models on WLASL because of their superior temporal modeling of images of signs and signals in asylum seekers referral centres [11].

6) Advanced training strategies: Recent studies emphasize the importance of training strategies such as frame sampling, data augmentation, and handling variable-length sequences. These factors have far-reaching implications in the real world sign language recognition system as Kvanchiani et al. emphasized that they influence performance considerably as well as sign language recognition systems in reality [12].

D. Summary

CNN-based models are highly accurate in static recognition but do not have a time sense. Hybrid models are better at modeling time, yet are not very scalable. Video-based architectures, including SlowFast, and transformer-based ones, including ViViT, have better spatio-temporal learning at the expense of increased computational complexity. This comparison warrants application of CNN based techniques in ASL alphabet recognition and video based model in word level recognition in this project.

IV. Project Approach

To develop an application design of Sign Language translator, the work was divided into 2 big parts: researching and evaluating different models for translation and developing the application. First part was further divided into several segments, as solutions for both ASL alphabet recognition using static images and video-based word-level recognition using dynamin video sequences were suggested.

A. Image Translation

1) Datasets: The study uses two datasets for static ASL imagery: ASL Alphabet and Images of American Sign Language (ASL) Alphabet Gestures.

ASL Alphabet dataset: The dataset is collection of images of alphabets from the American Sign Language. There are 29 classes, of which 26 are for the letters A-Z and 3 classes for 'space', 'delete' and 'nothing'. The images are 200x200 pixels. The training set contains 87000 images, while the test set contains 29 images (1 image per class) [5].

Images of American Sign Language (ASL) Alphabet Gestures: This dataset contains images of American Sign Language (ASL) gestures. It contains a total of 210,000 images of 28 classes, of which 26 are for the letters A-Z and 2 classes for 'space' and 'delete'. [6].

One of the main issues encountered in this part of the project was that the datasets had more than the number of images required for model training. Training the models on the entire dataset results in an accuracy greater than 99% for most of them. Therefore, comparing the models becomes meaningless. To address this problem, both datasets were downsampled to two sets with 30 and 300 images per class.

2) Models and Configuration: For this task, several CNN-based, RNN-based, Transformer-based, and hybrid models were considered. After some research, the following models and hybrids were chosen for training and testing on the available datasets: AlexNet, ResNet18, LSTM, Vision Transformer (ViT), AlexNet + LSTM, AlexNet + ViT, ResNet18 + LSTM, ResNet18 + ViT. Models were trained using Cross Entropy Loss and Adam optimizer.

B. Video Translation

1) Dataset: The study uses the WLASL-2000 Resized dataset [13]. It is the largest video dataset for Word-Level American Sign Language recognition, featuring 2000 common different words in ASL. The data set can be downsampled to three smaller sets: WLASL-100, WLASL-300, and WLASL-1000. Due to the limit in available resources, WLASL-100 and WLASL-300, containing videos of 100 and 300 different words in ASL, were used for model training and comparison. Model fine-tunings were implemented in the Kaggle platform with GPU T4 x2.

In addition, two custom simple datasets were created to test the models in 'real-life'-like environment. These two datasets, called 'Custom-100 and Custom-300, consist of the same 100 and 300 classes from WLASL-100 and WLASL-300. However, each class has only 1 video recorded by the participants of this project.

*2) **ViViT model**:* Video Vision Transformer (ViViT) is a pure-transformer model for video classification presented by Arnab et al [3]. For this model, the video must be mapped to a sequence of tokens. The paper describes several methods of embedding videos, one of them being the Uniform frame sampling method. It is a straightforward way of tokenizing the input video by uniformly sampling the required number of frames from the video and embedding each frame independently using the same method as ViT.

For inference, the input to their network is 4 video clips of 32 frames using a stride of 2. Then they averaged per-view logits to obtain the results. Using this method, they covered the duration for most of the videos in their dataset, which had around 250 frames on average.

The videos in WLASL dataset on average are around 70 fps. For training, it is possible to use the same uniform frame sampling method, so 32 frames required by the

model will be obtained that way. However, the video embedding method used for inference is not applicable in this case, therefore it was slightly altered. Instead of 4 clips with a stride of 2, 2 clips with a stride of 1 will be captured, which should be sufficient to cover the video duration.

At this stage one of the main issues was encountered: full-finetuning of the ViViT model pretrained on Kinetics-400 dataset required more time and resource (that were limited on Kaggle) than anticipated. Therefore, ViViT was pre-trained on 'Shawon16/ViViT_wlasl_100_200ep_coR_' from Hugging Face. Furthermore, instead of using predetermined number of epochs, Early Stopping with the patience of 5 (meaning if model does not improve in 5 epochs, fine-tuning stops) was implemented.

*3) **SlowFast and Slow-only Networks**:* SlowFast and Slow-only Networks: SlowFast networks proposed by Facebook AI Research Team can be considered as an extended two-stream method which operates at two different frame rates [4]. The idea behind this model lies in biological features of retinal ganglion cells in the primate visual system. Specifically, it consists of two types of cells such as Parvocellular (P-cells, $\sim$80%) and Magnocellular (M-cells, $\sim$15-20%). While P-cells function on low temporal frequency and capture spatial specific details, M-cell operates at higher rate and perceive fast temporal motion but less attentive to details. In the same way, Slow Pathway processes fewer frames with more channels to extract spatial semantics, while the Fast Pathway processes more frames with fewer channels to model motion patterns.

In our research, we use full fine-tuning on a pretrained on Kinetics-400 datasest SlowFast 8x8 model with ResNet-50 backbone. The training pipeline and data preprocessing were adapted from the official SlowFast repository [14], with pretrained weights loaded via the PyTorchVideo model zoo [15], where the model was pretrained on Kinetics-400. . The model name "8×8" means 8 Slow frames sampled with a temporal stride τ = 8. The Fast Pathway takes 32 frames as input, sampled from a random clip at sampling rate 2. The Slow Pathway takes 8 frames uniformly subsampled from those 32 frames, with α = 4 defining the ratio between the two pathways (sampling every 4th frame). While Slow Pathway operates at full channel capacity, Fast Pathway uses $\beta = \frac{1}{8}$ less channels operating at low channel capacity. Time-to-channel lateral connection method is used for fusion of Fast Pathway output to the Slow Pathway. The model takes 4 consecutive frames from Fast pathway and stacks them along channel dimension. In this way, the result is 1 output frame containing information from the original 4 frames. This process is repeated for 8 output time steps. Then reshaped Fast pathway features are projected via 3D Convolution to match the Slow pathway channel dimensions,and finally it is fused into Slow Pathway.

Frames sampling for training and validation:

- 32 frames sampled from a random clip at sampling rate = 2.
- Fast Pathway receives all 32 frames.
- Slow Pathway receives 8 (every 4th frame, α = 4) from fast frames.
- Random spatial cropping is applied.

The official implementation uses 10 uniform clips × 3 spatial crops (30 views). WLASL videos' average duration is only 2 - 3 seconds, while Kinetics' is 10 seconds. So, 10 clips would cause high overlapping of frames within the clips, it is reduced to 2. In addition, the sampling rate was reduced to 1 to cover two clips with lower overlapping of frames for prediction.

Frames sampling for testing:

- 2 uniformal clips with center crop is selected.
- 32 frames per clip at sampling rate = 1.
- Softmax scores averaged across clips for final prediction.

For the Slow-only comparison, we use full fine-tuning on a pretrained Slow 8x8 R50 model on Kinetics-400, operating only on the Slow Pathway with no lateral connections.

Frames sampling for training and validation:

- 8 frames sampled from a random clip at sampling rate = 8.

Frames sampling for testing:

- From each clip 8 frames are sampled at sampling rate = 4.

Normalization, optimizer, and augmentation settings remain same across both models. Training was conducted for 30 epochs.

V. Project Execution

The basic architecture is a comprehensive real-time sign language recognition system, which consists of two modes, namely, at the level of English letters, as well as words. At the initial stage, special attention was paid to static images rather than dynamic ones, recognizing gestures using a webcam in real time. To determine the best model, a comparison between AlexNet, ResNet, ViT, LSTM, as well as their hybrid models for effective and accurate recognition was carried out. After careful testing and evaluation, the best models were selected for both letter and word recognition. The first mode, due to its excellent reliability, showed decent results when working with a system that includes 29 classes, consisting of the English alphabet, "space", "delete" and "nothing" classes.During the development of the system,

various design and technical solutions were adopted. The OpenCV library allowed the basic architecture to use the webcam as an input; also it was used as a processing tool on the device, converting the received input into an output with corresponding text. The initial problem with the implementation was background noise, as well as a prediction based on the entire frame of the webcam output, which together reduced the accuracy of the prediction. It was decided to create a region of interest (ROI) that focused on right-hand gestures. Furthermore, to address frame-by-frame prediction instability ("flickering"), a time-based confirmation mechanism was introduced to stabilize outputs. The system logic was also refined to properly handle repeating letters, incorporating a "Current" and "Last Confirmed" feedback display to improve user interaction and output consistency. The next stage consisted of improving the efficiency of a system that could recognize sign language at the word level. Sequentially, the two modes were combined so that they could work together. A comparative analysis was performed for dynamic image processing between ViViT, Slow-only and SlowFast models. To optimize the performance, the dataset was filtered by eliminating poorly used signs and a regular temporal spacing policy was implemented so that the model always had a 32-frame input buffer needed to make proper inferences. In order to make a fair comparison and conclusion in terms of recognition performance and efficiency, 300 custom videos were created for each class to test the models. All the videos were 2 to 4 seconds long, and also had a speed of 30 frames per second (fps) to ensure consistency for the system when entering data. These technical improvements were reflected in the system interface, where the addition of a confidence indicator, word label (result), and the video output itself. A well-organized team structure was kept throughout the development process, where the tasks were split into two special subgroups: the team working on the development of the system and user interface integration, and the team working on the selection of the model, methodological design, and optimization. This concurrent workflow made it possible to make efficient advances in implementation of the application level as well as development of deep learning models, leading to a unified and complete system.

VI. Evaluation and Results

A. Image-based Model Results

1) Analysis: Based on the results shown in Table 2 I, ResNet is the most consistent model with the highest overall performance across both Dataset 1 and Dataset 2, and two training conditions (30 and 300 images/class). All models were evenly sampled so that each class contributed equal numbers of images to both training conditions, allowing for a fair comparison of performance.

TABLE I
Model accuracy comparison across Dataset 1 [5] and Dataset 2 [6]

Model	Dataset 1		Dataset 2	
	30 images	300 images	30 images	300 images
AlexNet	95.86	98.35	87.14	99.97
ResNet	97.93	99.87	93.57	99.64
LSTM	14.47	27.59	16.07	17.50
ViT	83.10	89.17	40.36	91.07
AlexNet+LSTM	86.90	98.82	66.79	98.93
ResNet+LSTM	97.24	99.65	89.64	99.29
AlexNet+ViT	90.83	96.83	86.43	98.21
ResNet+ViT	95.86	98.94	93.57	99.29

On Dataset 2, ResNet achieved the highest accuracy (97.93%) when trained with 30 images/class, exceeding AlexNet's 95.86% accuracy by 2.07% and outperforming ViT (83.10%) and LSTM (14.47%) significantly. In addition, ResNet achieved the highest accuracy (99.87%) when trained with 300 images, surpassing AlexNet (98.35%) by 1.52% and ViT (89.17%) by 10.70%. This indicates that ResNet scales well as data increases while also maintaining superior performance.

In Dataset 1, ResNet achieved the highest accuracy of 93.57% when trained with 30 images/class, which exceeded AlexNet (87.14%) and is significantly higher than both ViT (40.36%) and LSTM (16.07%). ResNet achieved 99.64% accuracy when trained with 300 images/class, which is only slightly less than AlexNet's 99.97% accuracy (0.33% lower) but ResNet remains very comparable. This small difference is not significant given ResNet performed consistently well in every other circumstance.

ViT and LSTM have shown very poor performance with limited amounts of training data (e.g., ViT = 40.36%, LSTM = 14.47% accuracy).

From the above results, ResNet model is only consistent model with accuracy >93% for all configurations. Based on this, ResNet is chosen as base model for real-time recognition with images.

2) Observations from System: From Figure 1 we can see that the letter gesture was recognized with 92.0% accuracy. ResNet model used results in high accuracy for real-time letter gesture recognition.

B. Video-based Model Results

Dataset: For video-based experiments WLASL100 and WLASL300 subsets of the WLASL dataset are used [7]. As can be seen from Table II, both subsets are splitted consistently for train/validation/test data with the ratio $\sim$70/$\sim$17/$\sim$13. And for additional testing custom dataset with 100 and 300 classes with 1 video for each was produced.

Fig. 1. Representation of the letter recognition system mode

TABLE II
DATASET DISTRIBUTION FOR WLASL100 AND WLASL300

Split	WLASL100		WLASL300	
	#Videos	%	#Videos	%
Train	1142	71	3549	70
Validation	338	16	901	17
Test	258	13	668	13
Total	**2038**	**100**	**5118**	**100**

1) Quantitative Results: The results of the comparison of all models on different datasets are collected in Table 2. SlowFast outperformed Slow-only in both subsets with a significant increase in performance from 1.04 % for WLASL100 (65.38% vs. 64.34%) to 6.14 % for WLASL300 (61.38% vs. 55.24%). ViViT experiments for 100 and 300 classes resulted in achieving 51.9% and 27.5% respectively. Eventually, the highest accuracy on both datasets was achieved by the SlowFast model (65.38% and 55.24%). Slow-only and SlowFast models show differences in training dynamics as seen in Table V. ViViT, SlowFast and Slow-Only models were found to achieve accuracies of 34%, 61%, and 53% respectively on the 100-class custom set. As for Custom-300 test set, ViViT - 14.67%, SlowFast - 55.33% and Slow - 38.67%. This performance gap expands beyond what was previously observed with the standard vocabulary size tests with SlowFast consistently producing better performance than either of the other model types.

2) Analysis: According to the theoretical results from papers, the most accurate model for training on Kinetics-400 dataset is ViViT model with 78.5% accuracy, followed by SlowFast (78%) and Slow-only(72.6%).

From practical experiments generally lower accuracies were achieved due to reduced training time and resource constraints.

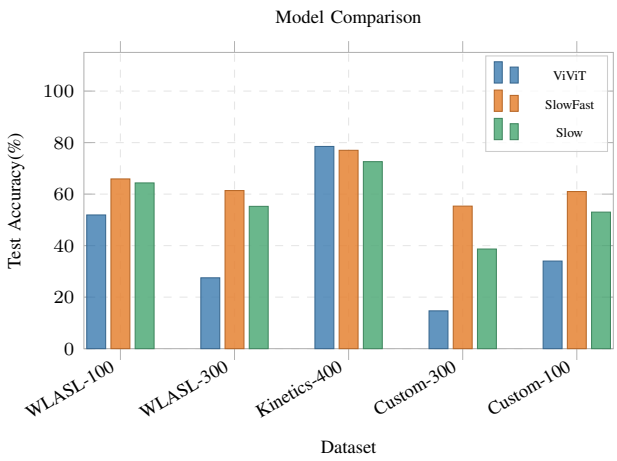


Fig. 2. Visual representation of the main results from Table III.

Each of the three models show decreasing performance from 100 classes to 300 classes, with SlowFast difference by 4.51%, Slow-only by 9.10%, and ViVit by 24.4%. The custom tests for one video for each class demonstrate an even more significant difference; SlowFast shows a 5.67% drop from 61% to 55.33%, Slow-only shows a 14.33% drop from 53% to 38.67%, and ViViT shows a 19.33% drop from 34% to 14.67%. Through this, we see that models that rely on a high level of spatio-temporal variation suffer significantly by not having sufficient amount of training information for each class.

There are two reasons affecting the reduce in accuracy. First is that deep learning models perform better on smaller datasets, which usually have less ambiguous data. With the increase of class number, more visually similar signs that are harder for classification tasks appear. Also, the difficulty of classification grows at least three times when using a vocabulary that grows larger than 100 classes while the number of training examples per class remains approximately constant across both subsets of data. This also restricts the ability to generalize from small to large classes in terms of accuracy [7].

Accuracy of SlowFast and Slow architectures for 100 classes differ only by 1.04% , while for 300 classes it reaches 6.14%. Such performance variance highlights the importance of dense temporal information for handling increased sign gestures recognition hardness. Using the Slow pathway alone results in 8 frames sampling with temporal stride 8, which does not capture fine-grained hand movements pattern. Adding Fast pathway leads to higher accuracy because of extracting features of 32 frames and fusing its spatial representation via lateral connections to Slow pathay. The training and validation gap shown in Table V proves the tendency of lower

TABLE III
MODEL PERFORMANCE COMPARISON ACROSS DATASETS (TEST ACCURACY %). KINETICS-400 VALUES ARE FROM [3], [4]. PRETRAIN SOURCES: SLOW-ONLY AND SLOWFAST PRETRAINED ON KINETICS-400; VIVIT PRETRAINED ON WLASL100 AND WLASL300. FOR CUSTOM DATASETS, MODELS ARE PRETRAINED ON CORRESPONDING WLASL SPLIT.

Model	Kinetics-400	WLASL-100	WLASL-300	Custom-100	Custom-300
Slow-only 8x8	72.6	64.34	55.24	53.00	38.67
SlowFast 8x8	77.0	65.38	61.38	61.00	55.33
ViViT	78.5	51.9	27.5	34.00	14.67

TABLE IV
ERROR VALUES OF SLOW-ONLY AND SLOWFAST MODELS ON WLASL DATASETS

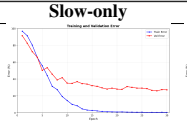
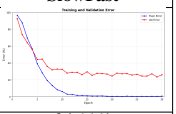
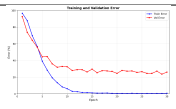
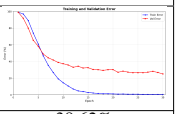
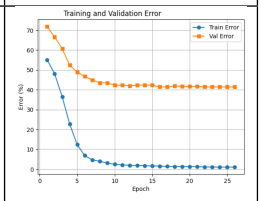
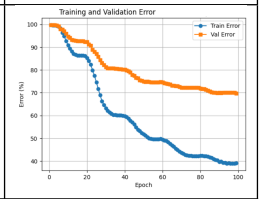
Dataset	Metric	Model	
		Slow-only	SlowFast
WLASL100	Train/Val Error		
	Test Error	35.66%	34.11%
WLASL300	Train/Val Error		
	Test Error	44.76%	38.62%

TABLE V
ERROR VALUES OF VIVIT MODEL ON WLASL DATASETS

Dataset	Metric	Model
		ViViT
WLASL100	Train/Val Error	
	Test Error	48.06%
WLASL300	Train/Val Error	
	Test Error	72.46%

generalization of the Slow model compared to SlowFast.

Contrary to expectations, ViViT performs worse than each of the two models, with results of 51.9% for WLASL100 and 27.5% for WLASL300 compared to SlowFast's results of 65.89% and 61.38%, respectively. The drop from WLASL100 to WLASL300 of 24.4% is significantly greater than that of SlowFast (4.51%) and Slow-only (9.1%). ViViT's mechanism for factoring spatio-temporally depends on the learning of relationships between a large number of training examples [3]. And during training with approximately ∼11% videos available for learning from per class, the model does not have enough diversity in the samples to be able to learn spatio-temporal patterns, which leads to it overfitting with 100 classes but failing to generalize with 300 classes due to insufficient samples diversity. Conversely, the SlowFast and Slow-only use inductive biases built into the convolution and thus remain effective even when the quantity of available data is low and also help maintain greater levels of consistency when looking at the two different groups.

From the above results, SlowFast is selected as best-performing model and is integrated in real-time word level recognition system.

3) Observations from System: From Figure 3 we can see that the word gesture was recognized with confidence score of 58.8%. SlowFast models used results in relatively high accuracy for real-time continuous gesture recognition.

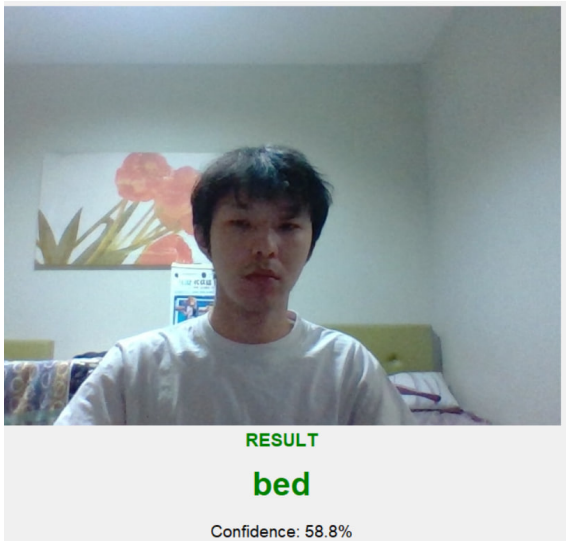


Fig. 3. Representation of the dynamic gesture recognition mode

VII. CONCLUSION AND FUTURE WORK

This project conducted evaluation on deep learning models for ASL recognition for static alphabet-level classification and dynamic word-level gesture

recognition. For image based recognition, ResNet architecture achieved the highest and consistent results ($> 93\%$), which validates the well-suitness of the CNN-based models for accurate static gestures recognition. For video-based recognition, SlowFast model showed better test accuracies (65.89%/61.38% on WLASL100/WLASL300) than Slow-only and SlowFast/ViViT models. The testing on custom test sets confirmed SlowFast model as best performing. From experimental results it is concluded that SlowFast Networks are effective for small datasets of sign gestures videos, and ViViT requires more training samples for the higher prediction values. In addition, the system can be improved by creating a user-friendly interface that looks concise and easy to manage. Moreover, when performing a certain gesture or low confidence prediction, a conditional video prompt with a specific gesture will pop up, which will show how to perform it correctly.

Although we introduced base pipelines of models and system design performing efficient recognition of alphabet- and word-level translation of gestures, several directions remain for future work. Firstly, this project can evolve in training on more diverse datasets with high quality images and videos. The second one is training ViViT and SlowFast models on WLASL with 1000 and 2000 classes for further accuracy evaluation gains. Furthermore, during this project, the methodology of input encoding for ViViT and SlowFast were simply adapted from [3] and [4]. Other methods of input encoding should be researched for possibly better results with WLASL dataset. For the system design, MediaPipe tool could be integrated for hand pose estimation for enhancing the robustness to background variation and lightning conditions [16].

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